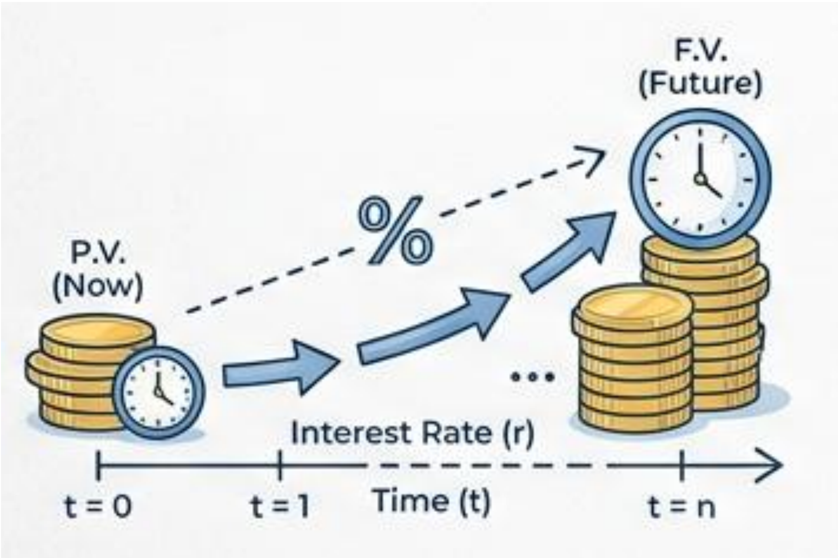
Introduction to Valuation



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Before You Come to Class — Required Reading

Read the **"Mr. Market"** section of Warren Buffett's 1987 Letter to Berkshire Hathaway Shareholders. It is only a few short paragraphs and is the most famous explanation of how to think about market prices.

Free online: www.berkshirehathaway.com/letters/1987.html (open the page, then use Find → "Mr. Market" to jump to the passage).

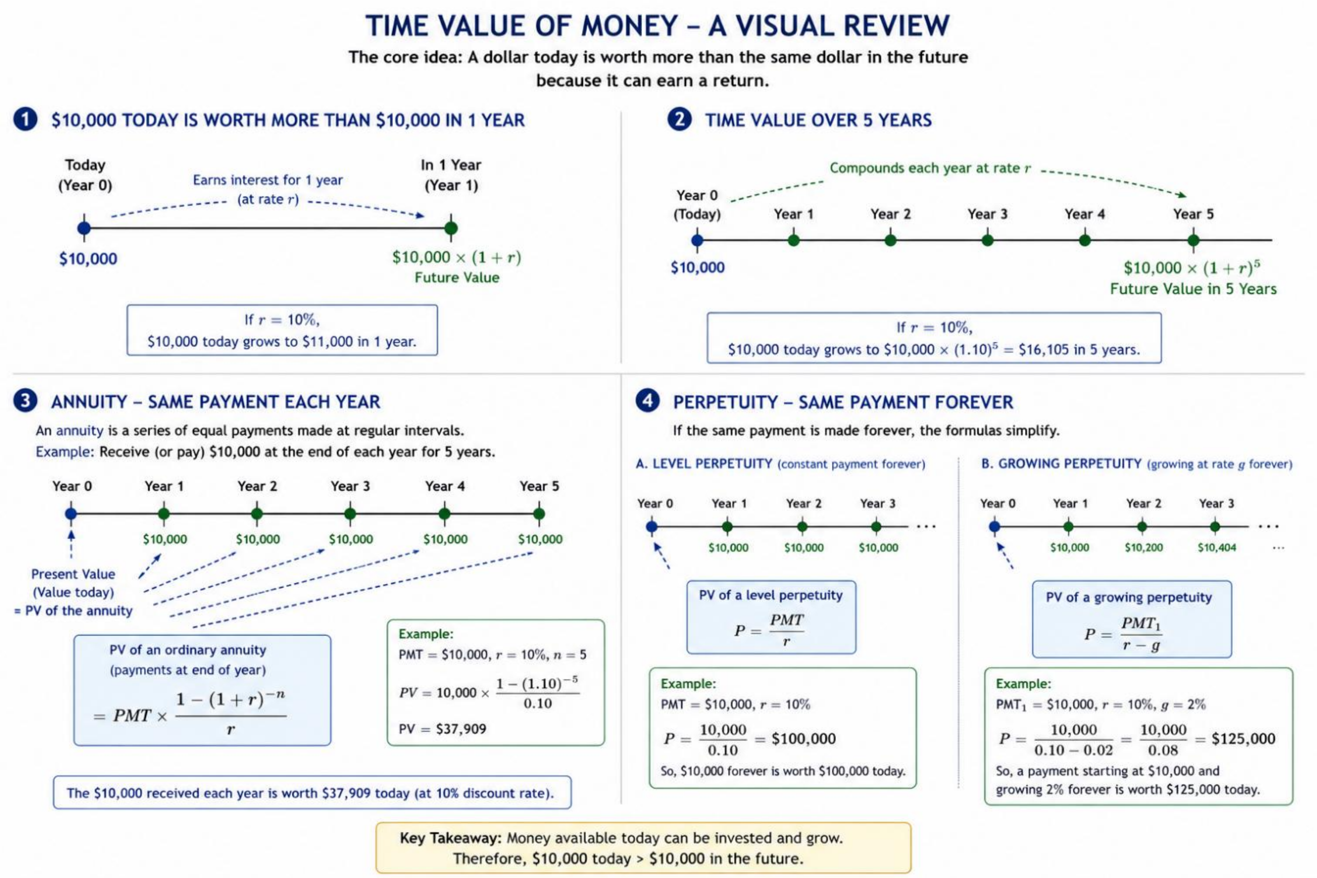
As you read, keep three questions in mind:

a. Who is Mr. Market, and what is "wrong" with him?
b. Why is he useful to a disciplined investor rather than dangerous?
c. What does Graham's "voting machine vs. weighing machine" idea mean?

Optional (for the curious): Benjamin Graham, *The Intelligent Investor*, Chapter 8 ("The Investor and Market Fluctuations") and Chapter 20 ("Margin of Safety").

Topic 1: The Time Value of Money -The Engine of Valuation

- A dollar today is worth more than a dollar in the future, because a dollar today can be invested to earn a return. This single idea — the **time value of money** — is the foundation of every valuation method in this course.
  - The interest rate (or **discount rate**), r, is the price of time. It converts dollars at one point in time into equivalent dollars at another.
- Two directions, one idea:
  - **Compounding** moves money forward in time (present → future).
  - **Discounting** moves money backward in time (future → present). Valuation is almost always discounting.



A. Future Value and Present Value of a Single Sum

- **Future value (FV)** — what a sum invested today will grow to:

$$FV = PV \times (1 + r)^n$$

- **Present value (PV)** — what a future sum is worth today (just FV rearranged):

$$PV = \frac{FV}{(1 + r)^n}$$

- Example: at r = 10%, $10,000 today grows to $11,000 in one year and to $16,105 in five years. Run in reverse: $16,105 to be received in five years is worth exactly $10,000 today.

B. Annuities — A Stream of Equal Payments

- An **annuity** is a series of equal payments made at regular intervals — for example, $10,000 at the end of each year for five years.
- PV of an *ordinary annuity* (payments at the *end* of each period):

$$PV = PMT \times \frac{1 - (1 + r)^{-n}}{r}$$

- Example: five annual payments of $10,000, discounted at 10%, are worth **$37,909** today — not $50,000 — because the later payments are discounted more heavily.

C. Perpetuities — A Stream That Never Ends

- A **perpetuity** is an annuity that continues forever. When payments last forever, the formulas collapse to something remarkably simple.
- **Level perpetuity** (constant payment forever):

$$P = \frac{PMT}{r}$$

  - Example: $10,000 per year forever at 10% is worth $100,000 today.
- **Growing perpetuity** (payment grows at rate g forever):

$$P = \frac{PMT_1}{r - g}$$

  - Example: a payment starting at $10,000 and growing 2% forever, discounted at 10%, is worth $125,000 today.
  - **Caution:** the formula only works when $r > g$, and PMT_1 is the cash flow *one period from today*.
- Why this matters most: the **growing perpetuity** is the single most important tool in valuation. In Topic 3 it reappears as the **terminal value** — the device we use to capture all cash flows beyond our explicit forecast horizon.

> **Key Takeaway:** Every valuation in this course is one rule applied repeatedly — discount expected future cash flows back to today at a rate that reflects their risk. Master discounting, and the rest is detail.

Concept Quiz #1

1. A dollar to be received three years from now is worth ____ a dollar received today (assume a positive interest rate).
   a. more than
   b. the same as
   c. less than
   d. cannot be determined
2. The present value of $5,000 received at the end of every year forever, discounted at 8%, is:
   a. $40,000
   b. $62,500
   c. $625,000
   d. infinite
3. Holding the payment and discount rate constant, increasing the growth rate g in a growing perpetuity will ____ its present value.
   a. increase
   b. decrease
   c. not change
   d. make it negative

Topic 2: Meet Mr. Market — Price vs. Value

- The next two topics will give us tools to estimate what a business is worth — its **intrinsic value**. **Mr. Market** is the parable that explains why the price the market quotes is so often different from that value.
- Benjamin Graham, in *The Intelligent Investor*, imagined the market as a business partner named Mr. Market. Warren Buffett retold the story in his 1987 letter to Berkshire Hathaway shareholders (your pre-class reading).

A. The Parable (in brief)

- Imagine you own a small business with a partner, Mr. Market. Every day, without fail, he offers to either buy your share or sell you his — and he names a price.
- Mr. Market is emotional. Some days he is euphoric and quotes a wildly high price; other days he is despondent and quotes a price far below the business's worth. His *mood*, not the business, drives his quote.
- The crucial point: Mr. Market is there to **serve** you, not to **instruct** you. You may trade with him when his price is attractive and ignore him when it is not. He will be back tomorrow with a new quote.

B. The Lesson: Price ≠ Value

- **Price** is what Mr. Market quotes today; **value** is what the business is actually worth. The two diverge constantly because price is set by crowd psychology in the short run.
- Graham's famous formulation: in the short run the market is a *voting machine*; in the long run, a *weighing machine*. Sentiment sets prices day to day; fundamentals set them over time.
- The investor's job is therefore not to predict Mr. Market's mood, but to (a) form an independent estimate of value, and (b) act only when price departs far enough from value to offer a **margin of safety** (Topic 8).

C. Two Ways People Try to Beat the Market

- **Fundamental analysis** — estimate intrinsic value from the business's cash flows and financial statements (the approach of this course), then buy when price is below value.
- **Technical analysis** — study patterns in past prices and volume to predict future prices. It ignores intrinsic value and focuses on Mr. Market's mood swings themselves.
- This course is firmly in the fundamental camp: we use accounting information to estimate value, and we treat market prices as *opportunities*, not instructions.

D. A Note on Market Efficiency

- The **efficient market hypothesis** holds that prices already reflect all available information, so Mr. Market is rarely "wrong." Reality is more nuanced: markets are largely efficient, but emotion, frictions, and information gaps create periodic mispricings.
- Whether or not you can consistently beat the market, the Mr. Market discipline — knowing your own estimate of value and refusing to be ruled by price — protects you from the most common and costly investing mistakes.

Key Takeaway: Mr. Market is your servant, not your guide. Use his quotes; don't be governed by them. The whole point of learning to value a business is to know when his price is a gift — and when it is a trap.

Discussion Questions

Required:

a. In your own words, explain the difference between the price of a stock and the value of the underlying business. Why do the two diverge?

b. Buffett writes that Mr. Market is there to serve you, not to instruct you. What does this imply about how an investor should react to a sudden 20% drop in a stock's price?

c. Graham distinguishes a voting machine from a weighing machine. Give one example (real or hypothetical) of each force acting on a stock price.

d. If markets were perfectly efficient, would the Mr. Market parable still be useful? Explain.

Topic 3: Discounted Free Cash Flow (DCF) Valuation

- The **DCF model** is the most complete expression of the time-value principle: the value of a business equals the present value of the **free cash flow** it will generate for its investors.
- Free cash flow (FCF) is the cash a business throws off *after* paying for the operations and investment needed to sustain it — the cash actually available to investors.
- FCF for Firm = CFO + Interest × (1 − t) − CapEx[1]
- We cannot forecast forever, so we split the future into two pieces:
  - An **explicit forecast horizon** (typically 5 years), where we project FCF year by year.
  - A **terminal value**, which captures the value of *all* cash flows beyond the horizon in a single figure.

$$\textit{Firm Value} = \textit{PV(FCF, Years 1–5)} + \textit{PV(Terminal Value)}$$

A. The Model, Stated Precisely

- The plain-English version above is exact. Here is the same idea in symbols, built in three steps — each step relaxing an assumption that the previous one could not carry.
- **Step 1 — firm value under certainty.** If every future cash flow were known with certainty, the value of the firm today would simply be the present value of all of them, discounted at *r*:

$$V_0 = \frac{FCF_1}{(1+r)^1} + \frac{FCF_2}{(1+r)^2} + \frac{FCF_3}{(1+r)^3} + \cdots + \frac{FCF_\infty}{(1+r)^\infty} \quad (1)$$

  - where V_0 is the value of the firm at $t = 0$, and FCF_n is the free cash flow at time $t = n$. This is Topic 1's discounting rule, applied to a stream that never ends.
- **Step 2 — firm value under uncertainty.** No one knows future cash flows. We replace each with its **expected value**:

$$V_0 = \frac{E(FCF_1)}{(1+r)^1} + \frac{E(FCF_2)}{(1+r)^2} + \frac{E(FCF_3)}{(1+r)^3} + \cdots + \frac{E(FCF_\infty)}{(1+r)^\infty} \quad (2)$$

  - where $E(\cdot)$ denotes the expected value. Note what has quietly changed: *r* must now compensate for risk as well as time, because the numerators are expectations rather than certainties. This is the risk force from the P/E topic, entering through the discount rate.
- **Step 3 — firm value with a finite forecast horizon.** Equation (2) is correct but unusable: forecasting to infinity is not possible. In practice analysts split the stream in two:
  - An **explicit forecast horizon** — typically five years — over which FCF is projected year by year.
  - A **terminal value** — a single figure capturing everything beyond the horizon.
- Equation (2) can then be rewritten as:

$$V_0 = \frac{E(FCF_1)}{(1+r)^1} + \cdots + \frac{E(FCF_5)}{(1+r)^5} + \frac{\textit{Terminal Value}}{(1+r)^5} \quad (3)$$

- which is exactly the two-piece formula stated above:

$$\textit{Firm Value} = \textit{PV(FCF, Years 1–5)} + \textit{PV(Terminal Value)}$$

- **What the three steps buy you.**
  - Nothing has been approximated between (1) and (3) **except** the assumption that the terminal value correctly summarizes the tail. The five-year horizon is not where the model becomes an estimate — the **expectations** in (2) already made it one.

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[1] Since CFO is computed after deducting interest, we add back interest net of tax — Interest × (1 − t) — to measure cash flow available to all capital providers, both debt and equity holders.

- The horizon is a matter of forecasting practicality, not theory. Five years is a convention, not a law.

B. The Five-Step Procedure

1. Forecast sales (revenue) for the explicit horizon, anchored to the most recent actual results.
2. Project expenses from the sales forecast to arrive at operating income.
3. Convert income into free cash flow (adjust for non-cash items and reinvestment).
4. Estimate the terminal value at the end of the horizon, assuming FCF grows at a steady long-run rate forever:

$$Terminal\ Value\ (end\ of\ Year\ n) = \frac{FCF_{n+1}}{r-g}$$

5. Discount the explicit FCFs and the terminal value to the present at the discount rate r, and sum them to obtain firm value.

C. A Worked Example

- Assume a discount rate r = 10% and a long-run growth rate g = 3%. The projected free cash flows and their present values are shown below ($ in millions).

Year	FCF ($M)	Discount factor @ 10%	PV of FCF ($M)
1	100.0	0.9091	90.9
2	115.0	0.8264	95.0
3	130.0	0.7513	97.7
4	145.0	0.6830	99.0
5	160.0	0.6209	99.3

Sum: PV of free cash flows (Years 1–5)	**$ 482.0**

- Now value everything beyond Year 5 with the growing-perpetuity (terminal value) formula:

FCF in Year 6 = $160.0M × (1 + 0.03)	$ 164.8
$\text{Terminal Value at end of Year 5} = \frac{\$164.8\text{M}}{0.10 - 0.03}$	$2,354.3
PV of Terminal Value = $2,354.3M × 0.6209	**$1,461.8**

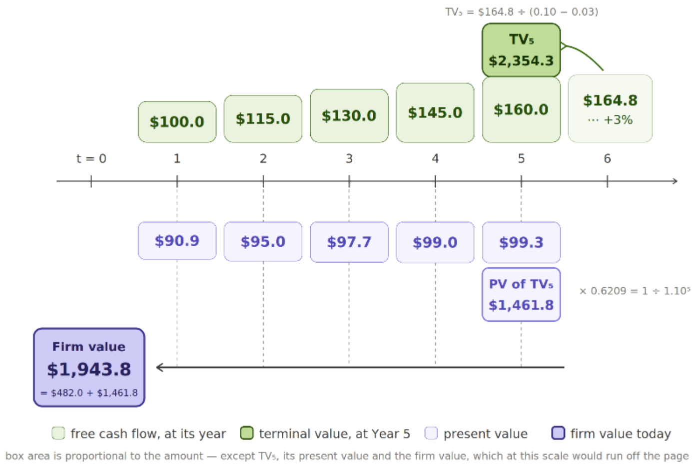
- Add the two present values to get the value of the whole firm, then bridge to value per share:

PV of free cash flows (Years 1–5)	$ 482.0
PV of terminal value	$ 1,461.8
Firm (enterprise) value	**$ 1,943.8**
Less: net debt	($ 300.0)
Equity value	**$ 1,643.8**
÷ Shares outstanding (millions)	100.0
Value per share	**$16.44**

D. The Punch Line on Terminal Value

- Notice the terminal value provides about **75%** of the firm's total value. This is typical — and it is why g and r deserve the most scrutiny: small changes in the spread (r − g) move the answer a lot.

- This also closes the loop with Topic 1 — the terminal value is simply a **growing perpetuity**, discounted back to today.



Key Takeaway: A DCF is only as good as its assumptions — garbage in, garbage out. Its real value is discipline: building one forces you to state, explicitly, what you believe about growth, profitability, reinvestment, and risk.

Concept Quiz #3

1. In a five-year DCF, the terminal value is used to capture:
   a. the cash flows of Years 1–5 only
   b. the value of all cash flows after the explicit forecast horizon
   c. the company's book value
   d. the dividends paid in Year 1
2. A firm's Year-6 FCF is expected to be \$210M and to grow at 3% forever; the discount rate is 9%. The terminal value at the end of Year 5 is:
   a. \$2,333M
   b. \$3,500M
   c. \$23,333M
   d. \$700M
3. Holding cash flows constant, narrowing the spread between r and g will cause the terminal value to:
   a. increase
   b. decrease
   c. stay the same
   d. become zero

Topic 4: The Cost of Equity

- Every valuation in this course discounts future cash flows back to today, and the **discount rate** is where risk enters the analysis. The DCF of Topic 3 is only as reliable as the rate we discount at — this topic explains where that rate comes from.
- The **cost of equity** is the return shareholders require for bearing the risk of owning a company's stock. It is the single most important — and most judgmental — input in equity valuation.

A. One Number, Three Names

- The same required return wears three hats, depending on who is looking at it. In an efficient market they are the **identical number**:
  - **Expected (required) return** — the investor's view: what shareholders expect to earn for the risk.
  - **Cost of equity capital** — the firm's view: what the company must deliver to attract equity. One person's return is another's cost.
  - **Discount rate** — the analyst's view: the rate that converts future equity cash flows into today's value.
- **Anchor:** a loan at 8% is at once the lender's expected return, the borrower's cost of borrowing, and the rate used to value the repayments — one rate, three seats.
- **Golden rule — match the rate to the cash flows**: discount equity cash flows (dividends, FCFE) at the **cost of equity**, and whole-firm cash flows (FCFF) at the **WACC**. Discounting equity cash flows at WACC double-counts the benefit of debt and overstates value.

B. The Building Blocks — CAPM

$$\textbf{Cost of Equity} = R_f + \beta \times ERP$$

- **Rf** — the risk-free rate: the *price of time*. Match its maturity to the cash-flow horizon — for going-concern equity use the 10-year Treasury, not the 3-month bill.
- **β (beta)** — how much the stock amplifies undiversifiable market risk; $\beta = 1$ is the market average.
- **ERP** — the equity risk premium: the *price of risk*.
- In words: expected stock return = pay for **waiting** (the risk-free rate) + pay for **worrying** (β × ERP).

C. The Equity Risk Premium

- Plain English: the extra return investors demand for holding risky stocks instead of safe government bonds — the market's "**hazard pay**." It is expected, not guaranteed; you earn it on average over the long run, not in every year.
- Two ways to estimate it:
  - **Historical** — the average realized gap between stock and bond returns (U.S. ≈ 5–6%). Backward-looking and constant.
  - **Implied** — solve for the return that sets today's index equal to the present value of expected cash flows (≈ 4–4.5% recently). Forward-looking and self-updating.
- A common working choice is about **5%**. There is no single "true" ERP — it is an assumption you disclose and defend.

D. Putting It Together — A Current Estimate

- With the 10-year Treasury near **4.6%** and an ERP of about **5%**, the cost of equity for an average (β = 1) firm is roughly **9.5%**.
- Scale by beta for other firms: a stock with β = 1.2 has a cost of equity of 4.6% + 1.2 × 5% ≈ **10.6%**.
- This r is exactly the **discount rate** you drop into the Topic 3 DCF — the two topics lock together: a valuation is only as trustworthy as the cost of equity behind it.

D. From Cost of Equity to the Cost of Capital (WACC)

- **Equity is only one source of financing.** Firms raise money from shareholders (equity) *and* lenders (debt), each demanding its own return. The cost of equity is what shareholders require; the **cost of debt** is what lenders require — roughly the firm's borrowing rate.
- **The firm's overall cost of capital blends the two — the WACC** (Weighted-Average Cost of Capital), weighting each by its share of financing:
  WACC = (weight of equity × cost of equity) + (weight of debt × after-tax cost of debt).
- **Debt is cheaper than equity** — lenders bear less risk, and interest is tax-deductible: after-tax cost of debt = cost of debt × (1 − tax rate). Some debt lowers the WACC; too much raises risk and pushes both costs back up.
- **Example.** Cost of equity 10%, cost of debt 5%, tax 20%, financing 60% equity / 40% debt. After-tax debt = 5% × (1 − 0.20) = 4%. **WACC = 0.60 × 10% + 0.40 × 4% = 7.6%.**
- **Which rate?** Discount shareholder cash flows (dividends, FCFE) at the **cost of equity**; discount whole-firm cash flows (FCFF) at the **WACC**.

Concept Quiz #4

1. The cost of equity, the return required by shareholders, and the discount rate for equity cash flows are best described as:
   a. three unrelated numbers
   b. the same required return viewed from three perspectives
   c. always equal to the risk-free rate
   d. numbers set by the government
2. A stock has a beta of 1.2. If the risk-free rate is 4.5% and the equity risk premium is 5%, its cost of equity under the CAPM is:
   a. 9.0% b. 9.5%
   c. 10.5% d. 6.0%
3. When you discount free cash flow to equity (FCFE), the appropriate discount rate is:
   a. the cost of equity
   b. the WACC
   c. the risk-free rate
   d. the cost of debt

Topic 5: The Price-Earnings (P/E) Ratio

- The **P/E ratio** is the market's most widely used valuation shorthand. It answers a simple question: how many dollars will investors pay today for one dollar of the company's annual earnings?

$$P/E = \frac{Market\ Price\ per\ Share}{Earnings\ per\ Share\ (EPS)}$$

  - Read it a **s a multiple** — a P/E of 15 means investors pay \$15 for every \$1 of current earnings.

A. A Quick Worked Example

- Company A earns EPS of \$4.00 and trades at \$60.00 per share:

Market price per share	\$60.00
Earnings per share (EPS)	\$ 4.00
P/E ratio	**15.0×**

- Run it in reverse to *value* a share: if comparable firms trade at 18× and Company A is expected to earn \$4.50 next year, a quick comp-based value is 18 × \$4.50 = **\$81.00** per share.

B. What Drives the P/E?

- Three forces push a P/E up or down — the same three that drive all value:
  - **Growth** — faster expected earnings growth → higher P/E.
  - **Risk** — higher risk (a higher discount rate) → lower P/E.
  - **Earnings quality** — more durable, cash-backed earnings → higher P/E (Topic 9).
- This is why a stable utility may trade at 12× while a fast-growing software firm trades at 40×. The market is not irrational; it is paying for growth.

C. The Algebra Behind the Three Forces

- The three forces are not a list to memorize. They all fall out of a single equation, and the starting point is the perpetuity from Topic 1.
- **Step 1 — no growth.** A firm that pays out all of its earnings and never grows is a perpetuity [2]:

$$P = \frac{E}{r} \quad so \quad P/E = \frac{1}{r}$$

  - This is the **risk** force standing alone: the P/E is simply the reciprocal of the discount rate. A higher r means a lower multiple, with nothing else in the picture.
- **Step 2 — let the firm reinvest.** Write the payout ratio as b, so the dividend is $D_1 = b \times E_1$. The Gordon growth model then gives:

[2] The value of a business is the present value of its **cash flows**, so a valuation formula strictly wants **free cash flow**, not accounting earnings. The two differ whenever there are non-cash charges or net reinvestment. **Step 1 is the one case where they coincide.** With no growth, net new investment is zero — capital expenditure merely replaces depreciation, and working capital is flat. So the items that normally separate earnings from free cash flow cancel out. **Earnings, dividends, and free cash flow are all equal here** — not as an approximation, but because the no-growth assumption forces it. That is what lets us write *E* in a formula that technically calls for cash flow.

$$P = \frac{b \times E_1}{r-g} \qquad so \qquad P/E = \frac{b}{r-g}$$

- Step 1 is the special case where g = 0 and b = 1. **Growth** enters by eating into the denominator — it does not act on its own, it acts on the spread between r and g.

▪ **Step 3 — let earnings quality matter.** If only a fraction q of reported earnings is durable and cash-backed, the market capitalizes q × E rather than E:

$$P = \frac{b \times E_1 \times q}{r-g} \qquad \text{So} \qquad P/E = \frac{b \times q}{r-g}$$

- **Earnings quality** sits in the numerator. Two firms with identical growth and risk will trade at different multiples if one firm's earnings are accrual-heavy (Topic 9).

▪ **Checking the 12× and the 40×.** The two firms from section B are not on the same curve:

- Utility: b = 0.6, q = 1, r = 7%, g = 2% → $\mathrm{P/E} = \frac{0.6}{0.05}$ = **12.0×**
- Software: b = 0.8, q = 1, r = 10%, g = 8% → $\mathrm{P/E} = \frac{0.8}{0.02}$ = **40.0×**
- The software firm carries the **higher** discount rate and still earns the higher multiple: narrowing the spread from 5% to 2% overwhelms the extra risk. That is the precise sense in which the market is "paying for growth."

▪ **Why a high multiple is a fragile multiple.** The spread is squared in the derivative, so a high P/E is a narrow spread — and a narrow spread is easily disturbed:

	g – 0.5 pp	**Base case**	**g + 0.5 pp**
Utility (r = 7%, g = 2%)	10.9×	**12.0×**	13.3× (+11%)
Software (r = 10%, g = 8%)	32.0×	**40.0×**	53.3× (+33%)

- The same half-point of growth moves the software multiple three times as far. A 40× multiple is not by itself a claim that the firm is expensive; it is a warning that small revisions to the assumptions will re-rate it sharply. This is why growth stocks fall so hard on an earnings miss.

D. Trailing vs. Forward P/E

▪ **Trailing P/E** uses the last twelve months of actual earnings; **forward P/E** uses next year's expected earnings.

Concept Quiz #5

1. A stock trades at $90 and reports EPS of $6.00. Its P/E ratio is:
   a. 6.0× b. 15.0×
   c. 0.067× d. 540×
2. All else equal, an increase in a firm's risk (its discount rate) should cause its justified P/E ratio to:
   a. increase b. decrease
   c. stay the same d. become negative

Topic 6: Free Cash Flow vs. Net Income as Valuation Input

- Topic 3 valued a firm by discounting its **free cash flow (FCF)**; the P/E ratio (Topic 5) values it off **earnings**. Are these two different theories of value? No — they are two routes to the same answer. This topic shows when you may use earnings in place of FCF, and what you must respect when you do.

A. Two Roads to the Same Value: DCF and the Residual Income Model

- The DCF model discounts expected free cash flows:

$$V_0 = \frac{\mathrm{FCF}_1}{(1+r)} + \frac{\mathrm{FCF}_2}{(1+r)^2} + \frac{\mathrm{FCF}_3}{(1+r)^3} + \cdots \qquad (1)$$

- With some algebra, the *same* value can be rewritten using earnings and book value — the **residual income (RI) model**:

$$V_0 = B_0 + \frac{\mathrm{RI}_1}{(1+r)} + \frac{\mathrm{RI}_2}{(1+r)^2} + \frac{\mathrm{RI}_3}{(1+r)^3} + \cdots \qquad (2)$$

  - where B_0 is today's book value of equity, and each period's residual income is earnings minus a charge for the cost of equity on the book value tied up in the business:

$$\mathrm{RI_t} = \mathrm{Net\ Income_t} - \mathrm{r} \times \mathrm{B_{t-1}}$$

- Equations (1) and (2) are **algebraically equivalent** — same value, different inputs. So valuing off net income is not a rival model; it is the DCF in different clothes.

B. The Constant-Growth Shortcut — FCF vs. Earnings

- If FCF grows at a constant rate g forever ($FCF_2 = FCF_1(1+g)$, $FCF_3 = FCF_2(1+g)$, …), the DCF collapses to the familiar perpetuity:

$$V_0 = \frac{\mathrm{FCF}_1}{r-g} \qquad (3)$$

- We can rewrite this in earnings. Because growth has to be *funded*, only part of earnings is actually distributable:

$$V_0 = \frac{E_1(1-b)}{r-g} \qquad (4)$$

- where E is earnings (net income) and **b is the plowback (retention) ratio** — the portion of earnings kept and reinvested in the firm to fund growth. The remaining (1 − b) is what actually reaches investors. Growth and retention are linked by g = b × ROE, so you cannot grow without retaining.
- **In practice, people often just use earnings in place of FCF. What justifies that?**
  - **Earnings smooth timing noise** — accrual accounting's matching principle makes earnings a steadier gauge of sustainable performance, while single-year FCF is thrown around by the timing of capex and working-capital swings (a growing firm can even show negative FCF while creating value).
  - **Still DCF, not a different model** — under clean-surplus accounting the earnings-based residual income model is algebraically equivalent to the FCF model; it reaches the same value by another route.
  - **Less weight on the terminal value** — residual income models anchor most of the value in today's book value plus near-term residual income, not in a speculative terminal value, so they are often more robust.
  - **They converge in steady state** — for a mature, no-growth firm, depreciation ≈ maintenance capex and net income ≈ FCF, so normalized earnings are a clean proxy.

- **Only if earnings are good quality** — this holds only when earnings are recurring and cash-backed; accrual-heavy or manipulated earnings make FCF the safer basis (ties to Earnings Quality, Topic 9).

- **A common shortcut — and its trap.** People often estimate next year's earnings and simply write

$$V_0 = \frac{E_1}{r-g} \qquad (5)$$

  - This quietly assumes the firm pays out *all* of its earnings (b = 0). But if nothing is retained, nothing funds growth — so g must be 0 as well, and (5) is really just the no-growth value $V_0 = E_1/r$. Used with $g > 0$ it **overstates value**, because it capitalizes earnings the firm must actually keep to grow. The honest version keeps the (1 − b) term, as in equation (4).

- **Caveat:** using earnings (and the P/E) is a shortcut, not a substitute for analysis. It breaks down when earnings are negative, volatile, or distorted by one-time items or aggressive accounting — exactly why earnings quality (Topic 9) matters. A clean DCF (Topic 3) remains the more complete tool.

Key Takeaway: Earnings and free cash flow are two doors to the same valuation. You may value off earnings — exactly, through the residual income model, or approximately, through E(1 − b)/(r − g) — as long as you respect the reinvestment (1 − b) that funds growth and trust the quality of the earnings. Capitalizing full earnings, E/(r − g), silently assumes the firm grows without reinvesting — which it cannot.

Concept Quiz #6

1. The residual income model and the free-cash-flow DCF model:
   a. usually give very different values
   b. are algebraically equivalent — the same value expressed with different inputs
   c. can only be used for firms that pay dividends
   d. ignore the cost of equity
2. In $V_0 = E_1(1 - b)/(r - g)$, the factor (1 − b) appears because:
   a. earnings must first be reduced for taxes
   b. part of earnings must be reinvested to fund growth, so only (1 − b) is paid out
   c. it adjusts earnings for inflation
   d. it converts net income into book value
3. Valuing a growing firm as $V_0 = E_1/(r - g)$ — full earnings, without the (1 − b) term — will generally:
   a. understate the value
   b. overstate the value, by crediting growth without charging for the reinvestment that funds it
   c. give exactly the right value
   d. work only if the discount rate is zero

Topic 7: Trailing vs. Forward P/E in Practice - SKHY & NVDA

- Topics 1–6 valued a business from its cash flows. Here we put the **P/E ratio** to work on two real AI-era stocks and read what the market is telling us through the gap between their trailing and forward multiples.
- Recall the two versions: **trailing P/E** divides today's price by the last twelve months of *actual* earnings; **forward P/E** divides it by the next twelve months of *expected* earnings.
- The relationship between the two is itself information. The price is the same numerator in both, so the only thing that changes is the earnings figure in the denominator:
  - forward P/E **<** trailing P/E ⇒ the market expects EPS to **rise** (a bigger denominator pulls the ratio down).
  - forward P/E **>** trailing P/E ⇒ the market expects EPS to **fall**.

$$\textit{Implied 1-Year EPS Growth} \approx \frac{\textit{Trailing P/E}}{\textit{Forward P/E}} - 1$$

A. The Snapshot — SK Hynix vs. NVIDIA

- Both companies sit at the center of the AI build-out: NVIDIA designs the accelerators, and SK Hynix supplies the **high-bandwidth memory (HBM)** that sits beside them. Here is how the market was pricing their earnings in late June 2026:

Metric	SK Hynix (KRX 000660)	NVIDIA (Nasdaq NVDA)
Trailing P/E (last 12 mo. actual EPS)	~17×	~32×
Forward P/E (next 12 mo. expected EPS)	~6×	~22×
Forward ÷ Trailing	~0.35	~0.69
Implied 1-year EPS growth*	**~+183%**	**~+45%**

**Implied growth ≈ (Trailing P/E ÷ Forward P/E) − 1. Figures are rounded snapshots from public market-data aggregators in late June 2026; multiples move daily.*

- Read the table in one sentence: *both* stocks have a forward P/E below their trailing P/E — so the market expects both to grow earnings — but SK Hynix's gap is far larger.

B. Why SK Hynix's Forward P/E Is So Much Lower Than Its Trailing P/E

- SK Hynix is a **cyclical memory maker** (DRAM and NAND) and the leading supplier of HBM for AI servers. Memory is essentially a commodity: prices and profits swing violently with the industry supply–demand cycle.
- The company swung from heavy **losses** in the 2023 memory downturn to **record profits** in the current AI-driven upcycle. Its trailing earnings still sit partway up that climb; its expected earnings are much higher.
- So a forward P/E of ~6× against a trailing P/E of ~17× is the market saying it expects SK Hynix's EPS to roughly **triple** over the next year:

SK Hynix trailing P/E	17×
SK Hynix forward P/E	6×
Implied 1-year EPS growth $= \frac{17}{6} - 1$	**+183%**

- In plain terms: at today's price, investors are paying only about *six times* next year's expected earnings — precisely because those earnings are expected to be far larger than last year's.

C. What the Low Forward P/E Means — Two Readings

- A low forward multiple on a cyclical company can mean two very different things. A careful analyst holds both in mind.
- **Reading 1 — genuine growth.** If the HBM and DRAM upcycle delivers the expected earnings jump, then ~6× forward earnings is simply price ÷ a much-bigger denominator, and the stock is reasonably priced or under-priced on forward numbers.
- **Reading 2 — the cyclical trap.** For a commodity producer, the *lowest* P/E often appears right at the **earnings peak** — when profits are highest and about to roll over. If the cycle turns, the optimistic forward EPS never arrives, and the "cheap" 6× multiple was a mirage.
  - This is the mirror image of a cyclical at the **bottom**, where depressed or negative earnings make the trailing P/E look huge even though the stock may be cheap.
- **The discipline:** for cyclical firms, judge value on **normalized (mid-cycle) earnings**, not a single peak-year forward figure — and insist on a margin of safety (Topic 8). Always ask whether the forward estimate simply assumes the up-cycle continues.

Source note: *P/E figures compiled from public market-data aggregators (e.g., stockanalysis.com, GuruFocus, Investing.com), late June 2026. Multiples are approximate, rounded, and change daily.*

D. The NVIDIA Contrast — Why the Absolute Multiples Differ

- NVIDIA also trades at a forward P/E below its trailing P/E (~32× down to ~22×), implying about +45% expected EPS growth — strong, but a far smaller step-down than SK Hynix:

NVIDIA trailing P/E	32×
NVIDIA forward P/E	22×
Implied 1-year EPS growth $= \frac{32}{22} - 1$	**+45%**

- The more striking difference is the **absolute** level: the market pays ~22× next year's earnings for NVIDIA but only ~6× for SK Hynix. Why nearly four times as much for a dollar of NVIDIA's forward earnings?
- **Earnings quality and durability**. NVIDIA is a design-and-software franchise — roughly 75% gross margins, a software ecosystem moat, and real pricing power — so its earnings are seen as more sustainable and less commodity-cyclical. SK Hynix sells a more commoditized product whose pricing is set by the memory cycle, so its profits are more volatile and viewed as less durable.
- **The lesson in one line:** the **gap** between trailing and forward P/E reflects the expected near-term growth *rate*; the **absolute** multiple reflects the market's view of earnings *quality*, durability, and risk. Two AI winners, very different multiples.

Key Takeaway: A forward and the trailing P/E below means the market expects earnings to grow — and the wider the gap, the faster the expected growth. But for a cyclical like SK Hynix, an unusually low forward multiple may signal either a real earnings surge or peak-cycle over-optimism, so confirm it against normalized earnings and a margin of safety. The absolute multiple (~6× vs. ~22×) reflects earnings quality, not just growth.

Concept Quiz #7

1. If a company's forward P/E is below its trailing P/E, the market is most likely expecting its earnings over the next year to:
   a. rise
   b. fall
   c. stay flat
   d. become negative
2. SK Hynix trades at roughly 17× trailing and 6× forward earnings. The implied one-year EPS growth is closest to:
   a. −65%
   b. +35%
   c. +95%
   d. +185%
3. NVIDIA's forward P/E (~22×) is far higher than SK Hynix's (~6×), though both benefit from AI demand. The best single explanation is that:
   a. NVIDIA's earnings are viewed as more durable and less cyclical
   b. SK Hynix is unprofitable
   c. NVIDIA pays no dividend
   d. SK Hynix has fewer shares outstanding

Topic 8: The Margin of Safety

- If Mr. Market explains *why* price and value diverge, the **margin of safety** tells you what to *do* about it. Graham argued these three words best capture sound investing.
- **Definition** — the margin of safety is the discount of purchase price to estimated intrinsic value. You buy a dollar of value for materially less than a dollar:

$$\textit{Margin of Safety} = \frac{\textit{Intrinsic Value} - \textit{Market Price}}{\textit{Intrinsic Value}}$$

- Its purpose is humility about error. Your valuation will be wrong to some degree — forecasts miss, discount rates are imprecise. Buying well below value gives you room to be wrong and still not lose money.

A. A Worked Example

- Suppose your DCF (Topic 3) estimates a stock's intrinsic value at $100 per share, and Mr. Market is quoting $65:

Intrinsic value (your DCF estimate)	$100.00
Market price (Mr. Market's quote)	$ 65.00
Margin of safety (in dollars)	$ 35.00
Margin of safety (as a %)	**35.0%**

- The 35% cushion means that even if your valuation were too optimistic by, say, 20% (true value ≈ $80), you would *still* own the business below its worth at the $65 you paid.

B. How Much Margin Is Enough?

- There is no universal number, but the principle scales with uncertainty:
  - Stable, predictable businesses with high earnings quality justify a **smaller** margin (perhaps 15–25%).
  - Volatile, hard-to-forecast, or lower-quality businesses demand a **larger** margin (often 40%+).
- The riskier and harder-to-value the business, the wider the discount you should insist on before buying.

C. Why It Connects Everything

- The margin of safety is where the whole class comes together: you estimate value with the time-value tools (Topics 2–7), you accept that Mr. Market's price will swing around that value (Topic 2), and you protect yourself by buying only at a meaningful discount — and only when the earnings beneath your estimate are trustworthy (Topic 9).

Key Takeaway: Margin of safety = buying value at a discount to protect against the inevitable errors in your own forecasts. It converts an uncertain valuation into a disciplined, survivable investment.

Concept Quiz #8

1. An analyst estimates a stock's intrinsic value at $50 and can buy it at $40. The margin of safety is:
   a. 10%
   b. 20%
   c. 25%
   d. 80%
2. The primary purpose of the margin of safety is to:
   a. guarantee a profit on every investment
   b. protect against errors in the analyst's valuation
   c. predict short-term price movements
   d. eliminate the need to value the business
3. All else equal, a business with highly uncertain future cash flows should require a margin of safety that is:
   a. larger
   b. smaller
   c. exactly zero
   d. negative

Topic 9: Earnings Quality

- Every valuation tool in this course runs on accounting numbers — earnings, cash flows, book values. **Earnings quality** asks the essential question: can we trust the reported numbers as a guide to the future?
- **High-quality earnings** are (a) *sustainable* (likely to recur), (b) *cash-backed* (not just accruals), and (c) a *faithful* picture of true economic performance.
- **Low-quality earnings** are the opposite — propped up by one-time gains, aggressive accounting choices, or accruals that reverse later. A P/E or DCF built on low-quality earnings is precise nonsense.

A. Why Earnings and Cash Can Diverge

- Accrual accounting records revenues when *earned* and expenses when *incurred* — not when cash changes hands. This is appropriate, but it gives management latitude (estimates, timing, classifications) that can be used to flatter results.
- A useful first screen: compare **net income** with **operating cash flow** over time. When earnings persistently and increasingly exceed operating cash flow, the gap is a red flag worth investigating.

B. Common Red Flags

Red flag	What it may signal
Net income rising while operating cash flow stagnates or falls	Earnings supported by accruals, not cash
Revenue growing far faster than receivables can be collected	Aggressive or premature revenue recognition
Heavy reliance on “non-recurring” items that keep recurring	Core earnings weaker than they appear
Frequent changes in estimates, reserves, or accounting policies	Results may be managed to hit targets
Large or growing gap between GAAP and “adjusted” earnings	Inconvenient costs excluded from the headline number

C. The Practical Discipline

- Before trusting an earnings figure for valuation, ask three questions: Is it *recurring*? Is it *cash-backed*? Would it survive a skeptical reading of the *footnotes*?
- Earnings quality is not mainly about catching fraud (though it sometimes does). It is about knowing how much weight a number can bear before you build a valuation on top of it.

Key Takeaway: Earnings quality is the trust you can place in the numbers. A valuation is only as reliable as the earnings beneath it — so vet the earnings before you discount them.

Concept Quiz #9

1. Which of the following is the strongest single indicator that reported earnings may be of low quality?
   a. Earnings that have grown steadily for a decade
   b. Net income that consistently and increasingly exceeds operating cash flow
   c. A dividend that rises each year
   d. A low P/E ratio
2. “High-quality” earnings are best described as earnings that are:
   a. as large as possible
   b. sustainable and backed by cash
   c. reported under pro-forma rather than GAAP rules
   d. growing faster than revenue
3. Why does earnings quality matter for a DCF or P/E valuation?
   a. It does not; only growth matters
   b. The valuation is only as reliable as the earnings it is built on
   c. It changes the discount rate to zero
   d. It eliminates the need for a terminal value

Appendix- Warren Buffett's Famous Investing Quotes

1. **"Price is what you pay. Value is what you get."**
   → Focus on intrinsic value, not just the stock price.

2. **"Rule No. 1: Never lose money. Rule No. 2: Never forget Rule No. 1."**
   → Protecting capital matters enormously.

3. **"Be fearful when others are greedy and greedy when others are fearful."**
   → Great opportunities can appear when markets are pessimistic.

4. **"Our favorite holding period is forever."**
   → Buy excellent businesses you'd be comfortable owning for decades.

5. **"It's far better to buy a wonderful company at a fair price than a fair company at a wonderful price."**
   → Business quality often matters more than getting the absolute cheapest v aluation.

Buffett's philosophy in one sentence: Buy high-quality businesses at sensible prices, understand what you own, avoid permanent losses, and let compounding work for a very long time.

-- The End of the Valuation Primer --